

DIRICHLET AND RESISTANCE FORMS

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ABSTRACT. This mostly follows [Kig01, Appendix B], which is a summary of parts of [FÖT94].
See [Eva08, Appendix A] for a summary of the material on Dirichlet Forms.

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1. Equivalence of Closed Forms, Non-Neg S.A. Operators, S.C. Semigroups

$\mathcal{H}$ is a real (or complex) Hilbert space with inner product $(\cdot, \cdot)$ and norm $\|\cdot\|$. Think of an L^2 space.

1.1. Non-Negative Self Adjoint Operators.

Definition 1.1. $H : \mathcal{H} \rightarrow \mathcal{H}$ is a *non-negative self adjoint operator* if

- (1) $\text{Dom}(H) \subset \mathcal{H}$ is a dense subspace;
- (2) H is linear and symmetric, i.e.

$$(Hf, g) = (f, Hg) \quad \forall f, g \in \text{Dom } H;$$

- (3) H is non-negative, i.e.

$$(Hf, f) \geq 0 \quad \forall f \in \text{Dom}(H);$$

- (4) H is self-adjoint, i.e. $H = H^*$, i.e. (where h is denoted H^*g)

$$\text{Dom}(H) = \left\{ g : \exists h \in \mathcal{H} \text{ s.t. } (Hf, g) = (f, h) \quad \forall f \in \text{Dom}(H) \right\}.$$

Proposition 1.2. *If H is non-negative self adjoint operator then there exists a unique non-negative self adjoint operator G (denoted $H^{1/2}$) such that*

$$\text{Dom } H = \{f \in \text{Dom } G : Gf \in \text{Dom } G\}, \quad G^2 = H.$$

Think of $H = -\Delta$, $\mathcal{H} = L^2(\mathbb{R}^n)$ or $L^2(\Omega)$, $\text{Dom } H = W^{2,2}(\mathbb{R}^n)$ or $W_0^{2,2}(\Omega)$, $\text{Dom } H^{1/2} = W^{1,2}(\mathbb{R}^n)$ or $W_0^{1,2}(\Omega)$.

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1.2. Closed Forms.

Definition 1.3. $Q : \text{Dom}(Q) \times \text{Dom}(Q) \rightarrow \mathbb{R}$ is a (*non-negative quadratic*) *closed form* if

- (1) $\text{Dom}(Q) \subset \mathcal{H}$ is a dense subset;
- (2) Q is bilinear and symmetric;
- (3) Q is non-negative definite;
- (4) $\text{Dom}(Q)$ is a Hilbert space under the inner product

$$Q_*(f, g) := Q(f, g) + (f, g).$$

(or equivalently, $\text{Dom}(Q)$ is closed in $\mathcal{H}$ under the Q_* metric).

1.3. Strongly Continuous Semigroups.

Definition 1.4. $(T_t)_{t \geq 0}$ is a strongly continuous semigroup on $\mathcal{H}$ if

- (1) $(T_t)_{t \geq 0}$ are *bounded* linear symmetric (hence self adjoint) operators on $\mathcal{H}$;
- (2) $T_{t+s} = T_t \circ T_s \ \forall t, s > 0$ (i.e. $(T_t)_{t \geq 0}$ is a semigroup);
- (3) $\lim_{t \rightarrow 0^+} \|T_t u - u\| = 0$ (strong continuity at 0);
- (4) $\|T_t u\| \leq \|u\| \ \forall u \in H, t > 0$ (contractivity).

1.4. Equivalence of These Notions.

Theorem 1.5. *There is a natural one-one correspondence between non-negative quadratic closed forms Q , non-negative self adjoint operators H , and strongly continuous semigroups $(T_t)_{t \geq 0}$.*

For $Q \iff H$:

$$\text{Dom}(Q) = \text{Dom}(H^{1/2}) \supset \text{Dom}(H) = \{f \in \text{Dom}(H^{1/2}) : H^{1/2}f \in \text{Dom}(H^{1/2})\},$$

and

$$\begin{aligned} Q(f, g) &= (f, Hg) \quad \forall f \in \text{Dom}(Q), \forall g \in \text{Dom}(H), \\ &= (H^{1/2}f, H^{1/2}g) \quad \forall f, g \in \text{Dom } Q. \end{aligned}$$

In particular, Q determines H uniquely and conversely, by density of domains.

For $H \iff (T_t)_{t \geq 0}$:

The generator of $(T_t)_{t \geq 0}$ equals $-H$, i.e.

$$-Hf = \lim_{t \rightarrow 0^+} \frac{T_t f - f}{t},$$

where the limit is in the strong ($\|\cdot\|$) sense, and $\text{Dom}(H)$ is the set of f for which the limit exists. This gives H in terms of T_t .

Conversely, there is a unique solution of¹

$$u(0) = \phi \ (\in \mathcal{H}), \quad \frac{du}{dt} = -Hu \text{ for } t > 0,$$

and it is given by $u(t) = T_t \phi$.

It follows from the uniqueness of solutions that

$$H\phi = \lambda\phi \implies T_t \phi = e^{-\lambda t} \phi.$$

The following is straightforward to prove, see [Kig01, p200].

Proposition 1.6. *If $f, g \in \text{Dom } H$ then*

- (1) $-HT_t f = -T_t Hf = \frac{d}{dt} T_t f$ (limit in the strong sense);
- (2) $Q(T_t f, g) = Q(f, T_t g)$ for $t > 0$, $Q(T_s f, T_t g) = Q(T_t f, T_s g) = Q(f, T_{t+s} g)$ for $s, t > 0$;
- (3) $Q(T_t f, T_t f)$ is monotonically decreasing on $(0, \infty)$, $Q(T_t f, T_t f) \uparrow Q(f, f)$ as $t \downarrow 0$.

¹Where it is required that $u(t)$ is strongly continuous in t for $t \geq 0$, $u(t) \in \text{Dom}(H)$ for $t > 0$, and the derivative is taken in the strong sense.

2. Dirichlet Forms

Suppose X is a compact metric space and μ is a finite measure on X . Suppose $\mu(U) > 0$ for every nonempty open set U .

Let $C(X)$ be the set of continuous real-valued functions on X .

Suppose $\mathcal{E}$ is a closed form on $L^2(X, \mu)$ with $\text{Dom } \mathcal{E} = \mathcal{F}$.

Definition 2.1. Suppose $\mathcal{E}$ is a closed form on $L^2(X, \mu)$ with $\text{Dom } \mathcal{E} = \mathcal{F}$.

- (1) $(\mathcal{E}, \mathcal{F})$ is *Markovian* if the unit contraction operates on $\mathcal{E}$, i.e.:

$$f \in \mathcal{F} \implies \bar{f} := (f \wedge 1) \vee 0 \in \mathcal{F} \text{ and } \mathcal{E}(\bar{f}) \leq \mathcal{E}(f).$$

We then say $(\mathcal{E}, \mathcal{F})$ is a *Dirichlet form*.

- (2) $\mathcal{C} \subset \mathcal{F} \cap C(X)$ is a *core* of $(\mathcal{E}, \mathcal{F})$ if

(a) $\mathcal{C}$ is dense in $\mathcal{F}$ in the $\mathcal{E}_*$ norm (see Definition 1.3(4)),

(b) $\mathcal{C}$ is dense in $C(X)$ in the sup norm.

If $(\mathcal{E}, \mathcal{F})$ is a Dirichlet form with a core then it is called a *regular Dirichlet form*.

- (3) $(\mathcal{E}, \mathcal{F})$ has the *local property* if

$$\text{spt } f \cap \text{spt } g = \emptyset \implies \mathcal{E}(f, g) = 0.$$

If $(\mathcal{E}, \mathcal{F})$ is a Dirichlet form with the local property it is called a *local Dirichlet form*.

Definition 2.2. A strongly continuous semigroup $(T_t)_{t>0}$ on $L^2(X, \mu)$ is *Markovian* if

$$0 \leq f \leq 1 \text{ a.e.} \implies 0 \leq T_t f \leq 1 \text{ a.e. } \forall t > 0.$$

Theorem 2.3. A Dirichlet form is Markovian iff the corresponding semigroup as in Theorem 1.5 is Markovian.

3. Dirichlet Forms on Finite Sets

See [Kig01, §2.1]

Notation 3.1.

- (1) V is a *finite* set.
- (2) $\ell(V) = \{f \mid f : V \rightarrow \mathbb{R}\}$.
- (3) $(u, v) = \sum_p u(p)v(p)$.
- (4) For $u \in \ell(V)$,

$$\bar{u}(p) = \begin{cases} 1 & u(p) \geq 1 \\ u(p) & 0 \leq u(p) \leq 1 \\ 0 & u(p) \leq 0 \end{cases}$$

The notion of a Dirichlet form in Definition 2.1 reduces to the following (except for the characterisation of the kernel):

Definition 3.2. A symmetric non-negative bilinear form $\mathcal{E}$ on $\ell(V)$ is a *Dirichlet form* provided

- (1) $\mathcal{E}(u) = 0$ iff u is constant,
- (2) $\mathcal{E}(\bar{u}) \leq \mathcal{E}(u)$. (The *Markov* property.)

Example 3.3. Let $V = \{p_1, p_2, p_3\}$, set $u_i = u(p_i)$ and define

$$\mathcal{E}(u) = (u_1 - u_2)^2 + (u_2 - u_3)^2 + \epsilon(u_1 - u_3)^2.$$

Then $\mathcal{E}$ is a Dirichlet form for $\epsilon \geq 0$, while for $\epsilon > -1/2$ it satisfies all conditions except the Markov property.

The notion of a non-negative self adjoint operator in Definition 1.1 reduces to the following (except for the sign change and the characterisation of the kernel):

Definition 3.4. A symmetric non-positive linear operator (i.e. matrix) H on $\ell(V)$ is a *laplacian* provided

- (1) $H(u) = 0$ iff u is constant, (so rows sum to 0),
- (2) $H_{pq} \geq 0$ if $p \neq q$.

The following is a special case of Theorem 1.5.

Proposition 3.5. *There is a natural bijection between the set of Laplacians H and the set of Dirichlet forms $\mathcal{E}_H$, given by*

$$\mathcal{E}_H(u, v) = -(u, Hv) = -\sum_{p,q} H_{pq} u_p u_q = \frac{1}{2} \sum_{p,q} H_{pq} (u_p - u_q)^2.$$

Proof. The main point is that $\sum_{p,q} H_{pq} u_p^2 = 0 = \sum_{p,q} H_{pq} u_q^2$ by the first condition in Definition 3.4. $\square$

Remark 3.6. There is a natural electrical network induced by H or equivalently $\mathcal{E}$, with *resistance* $r_{pq} := H_{pq}^{-1}$ and *conductance* H_{pq} between p and q for $p \neq q$.

We say p is *connected to* q if $H_{pq} > 0$, and V is connected if any two vertices can be joined by a path of connected pairs. $\square$

If $U \subset V$ then there is a *natural restriction* $\tilde{H}$ of H , or equivalently $\tilde{\mathcal{E}}$ of $\mathcal{E}$, to U . We often think of U as the “boundary” of V . We will see that $\tilde{H}$ and H have the same effective resistance on U .

Theorem 3.7. *Suppose H is a Laplacian on V with corresponding Dirichlet form $\mathcal{E}$. Suppose $U \subset V$. For $u \in \ell(V)$ we write $u = \begin{bmatrix} u_0 \\ u_1 \end{bmatrix}$ where $u_0 = u|_U$ and $u_1 = u|_{V \setminus U}$.*

Let

$$H = \begin{bmatrix} T & {}^t J \\ J & X \end{bmatrix}$$

where the vertices in U are enumerated first and then those in $V \setminus U$. Thus $T : \ell(U) \rightarrow \ell(U)$, $J : \ell(U) \rightarrow \ell(V \setminus U)$ and $X : \ell(V \setminus U) \rightarrow \ell(V \setminus U)$.

Then

$$(1) \quad \mathcal{E}_H(u) = \mathcal{E}_{\tilde{H}}(u_0) + \mathcal{E}_X(u_1 + X^{-1}Ju_0),$$

where $\tilde{H} := T - {}^t JX^{-1}J$ and X are Dirichlet forms (equivalently, Laplacians) on U and $V \setminus U$ respectively.

If $h \in \ell(V)$ is defined by $h_0 = u_0$ and $h_1 = -X^{-1}Ju_0$, then h is the H -harmonic extension of u_0 in that $(Hh)_{V \setminus U} = 0$. Moreover,

$$\mathcal{E}_{\tilde{H}}(u_0) \leq \mathcal{E}_H(u)$$

and equality holds iff u is the H -harmonic extension of u_0 .

Proof. For (1), one just “completes the square”. In the case T , J and X are scalars this is

$$\begin{aligned} \langle Hu, u \rangle &= Tu_0^2 + Ju_1u_0 + Ju_0u_1 + Xu_1^2 \\ &= Tu_0^2 + X(u_1^2 + 2X^{-1}Ju_0u_1) \\ &= (T - X^{-1}J^2)u_0^2 + X(u_1 + X^{-1}Ju_0)^2. \end{aligned}$$

In general, and similarly,

$$\langle Hu, u \rangle = \langle (T - {}^t JX^{-1}J)u_0, u_0 \rangle + \langle X(u_1 + X^{-1}Ju_0), u_1 + X^{-1}Ju_0 \rangle.$$

The rest of the proof is straightforward. $\square$

Proposition 3.8 (Maximum Principle). *Suppose V is connected, H is a Laplacian on V and $U \subset V$. Suppose $u \in \ell(V)$ is H -harmonic on $V \setminus U$. Then for any $p \in V \setminus U$,*

$$\min_{q \in U} u(q) \leq u(p) \leq \max_{q \in U} u(q).$$

Each inequality is strict unless u is constant on U .

Proof. Standard. $\square$

Proposition 3.9 (Harnack Inequality). *Suppose V is connected, H is a Laplacian on V , $U \subset V$ and $A \subset V \setminus U$.*

Then there exists $c > 0$ such that if $u \geq 0$ on V , and u is H -harmonic on $V \setminus U$, then

$$\max_{p \in A} u(p) \leq c \min_{p \in A} u(p).$$

Proof. Let

$$\mathcal{A} = \{u \in \ell(V) : u \geq 0 \text{ on } V, u \text{ is } H\text{-harmonic on } V \setminus U, \max u|_A = 1\}.$$

Then $\mathcal{A} \subset \ell(V) = \mathbb{R}^d$ for $d = \text{card}(V)$ is compact, $\min_{p \in A} u(p) > 0$ for each $u \in \mathcal{A}$ by the maximum principle, and so $\gamma = \inf_{u \in \mathcal{A}} \min_{p \in A} u(p) > 0$. Let $c = \gamma^{-1}$. $\square$

Definition 3.10. For H a Laplacian on V with Dirichlet form $\mathcal{E}_H$ and $p \neq q$, the *effective resistance* is defined by

$$R_H(p, q)^{-1} = \min_{u \in \ell(V)} \frac{\mathcal{E}_H(u)}{|u(p) - u(q)|^2}.$$

In particular,

$$|u(p) - u(q)|^2 \leq R_H(p, q) \mathcal{E}_H(u).$$

(The minimum is achieved since w.l.o.g. we can assume $u(p) = 1$ and $u(q) = 0$, in which case we are minimising a quadratic over a finite dimensional space. The minimum is positive since $\mathcal{E}_H(u) = 0$ iff u is constant.)

Definition 3.11. Suppose $V_1 \subset V_2$ are finite sets with Laplacians H_1 and H_2 respectively. Then we write $(V_1, H_1) \preceq (V_2, H_2)$ if $H_1 = \widetilde{H_2}$ as in Theorem 3.7 and say the two networks are *compatible*.

Theorem 3.12. Suppose $V_1 \subset V_2$ are finite sets with Laplacians H_1 and H_2 respectively. Then $(V_1, H_1) \preceq (V_2, H_2)$ iff $R_{H_1}(p, q) = R_{H_2}(p, q)$ for all $p, q \in V_1$.

Proof. The $\implies$ direction is immediate.

The converse follows from the fact that a Laplacian is determined by its associated effective resistances. This latter is by a somewhat tedious induction on the cardinality of V . $\square$

Theorem 3.13. Suppose H is a Laplacian on V and R_H is the associated effective resistance. Then R_H is a metric.

Proof. Use the $\Delta - Y$ transform. $\square$

4. Sequences of Discrete Laplacians

See [Kig01, §2.2]

Notation 4.1.

- (1) $\mathcal{S} = \{(V_m, H_m)\}_{m \geq 0}$ is a *compatible sequence*, i.e. $(V_m, H_m) \preceq (V_{m+1}, H_{m+1})$ for $m \geq 0$.
- (2) $V_* := \bigcup_{m \geq 0} V_m$, $\ell(V_*) = \{f : V_* \rightarrow \mathbb{R}\}$.
- (3) $\mathcal{F}(\mathcal{S}) := \{u : V_* \rightarrow \mathbb{R} \mid \lim_{m \rightarrow \infty} \mathcal{E}_m(u|_{V_m}) < \infty\}$.
- (4) $\mathcal{E}_{\mathcal{S}}(u, v) := \lim_{m \rightarrow \infty} \mathcal{E}_{H_m}(u|_{V_m}, v|_{V_m})$ for $u, v \in \mathcal{F}(\mathcal{S})$.
- (5) The *effective resistance* associated with $\mathcal{S}$ is given by

$$R_{\mathcal{S}}(p, q) := R_{H_m}(p, q) \text{ if } p, q \in V_m.$$

Lemma 4.2. There exist unique $h_m : \ell(V_m) \rightarrow \mathcal{F}(\mathcal{S})$ satisfying

- (1) $h_m(u)|_{V_m} = u$;
- (2) $\mathcal{E}_{H_m}(u) = \mathcal{E}_{\mathcal{S}}(h_m(u)) = \min\{\mathcal{E}_{\mathcal{S}}(v) : v \in \mathcal{F}(\mathcal{S}), v|_{V_m} = u\}$, and the unique minimiser is $v = h_m(u)$;
- (3) If $n > m$ then $h_m(u)|_{V_n}$ is the harmonic extension of $u|_{V_m}$ to V_n .

Proof. Straightforward. $\square$

Lemma 4.3 (Maximum Principle). If $v \in \ell(V_*)$ is H_n harmonic on $V_n \setminus V_m$ for all $n > m$ then

$$\min_{q \in V_m} v(q) \leq v(p) \leq \max_{q \in V_m} v(q)$$

for all $p \in V_*$.

Proof. Straightforward. $\square$

Proposition 4.4. $R_{\mathcal{S}}$ is a metric on V_* .

Proof. Straightforward from the corresponding property for the R_{H_m} . $\square$

Lemma 4.5. For $u \in \mathcal{F}(\mathcal{S})$ and $p, q \in V_*$,

$$R_{\mathcal{S}}(p, q)^{-1} = \min \left\{ \frac{\mathcal{E}_{\mathcal{S}}(u)}{|u(p) - u(q)|^2} : u \in \mathcal{F}(\mathcal{S}), u(p) \neq u(q) \right\},$$

$$|u(p) - u(q)|^2 \leq R_{\mathcal{S}}(p, q) \mathcal{E}_{\mathcal{S}}(u).$$

Proof. Straightforward. $\square$

Proposition 4.6.

- (1) If $u \in \mathcal{F}(\mathcal{S})$ then u is uniformly continuous w.r.t. the metric $R_{\mathcal{S}}^{1/2}$.
- (2) $\mathcal{E}_{\mathcal{S}}$ is a non-negative symmetric bilinear form on $\mathcal{F}(\mathcal{S})$ and $\mathcal{E}_{\mathcal{S}}(u) = 0$ iff u is constant.
- (3) $(\mathcal{F}(\mathcal{S})/\sim, \mathcal{E}_{\mathcal{S}})$ is a Hilbert space, where $u \sim v$ iff $u - v$ is constant.
- (4) If $u \in \mathcal{F}(\mathcal{S})$ then $\bar{u} \in \mathcal{F}(\mathcal{S})$ and $\mathcal{E}_{\mathcal{S}}(\bar{u}) \leq \mathcal{E}_{\mathcal{S}}(u)$.

Proof. Straightforward except for (3). For this, identify $\mathcal{F}(\mathcal{S})/\sim$ with $\mathcal{F}_p(\mathcal{S}) = \{u \in \mathcal{F}(\mathcal{S}) : u(p) = 0\}$ for some fixed $p \in V_*$.

To show a Cauchy sequence $(v_k)_{k \geq 1}$ from $\mathcal{F}_p(\mathcal{S})$ has a limit in $\mathcal{F}_p(\mathcal{S})$, consider the sequence $v_k^m := h_m(v_k|_{V_m})$. This has a limit v^m and there exists $v \in V_*$ with $v|_{V_m} = v^m$ for all m . Moreover, $v \in \mathcal{F}(\mathcal{S})$ and one can check that the original sequence converges to v . $\square$

5. Resistance Forms and Resistance Metrics

See [Kig01, §2.3]

Beginning with a compatible sequence $\mathcal{S} = \{(V_m, H_m)\}$, we constructed in Notation 4.1 the “resistance” form $\mathcal{E}_{\mathcal{S}}$ on $\mathcal{F}(\mathcal{S})$ and the resistance metric $R_{\mathcal{S}}$ on V_* . Motivated by the properties in Proposition 4.4, Lemma 4.5 and Proposition 4.6 we define a general notion of “resistance metric” and “resistance form” and prove their equivalence.

Definition 5.1 (Resistance form). A pair $(\mathcal{E}, \mathcal{F})$ is a *resistance form* on an arbitrary set X if

- (1) $\mathcal{F}$ is a subspace of $\ell(X)$ which contains the constant functions.
- (2) $\mathcal{E}$ is a non-negative symmetric form on $\mathcal{F}$ and $\mathcal{E}(u) = 0$ iff u is constant.
- (3) $(\mathcal{F}/\sim, \mathcal{E})$ is a Hilbert space, where $u \sim v$ iff $u - v$ is a constant function.
- (4) For any finite $V \subset X$ and $v \in \ell(V)$ there exists an extension of v to some $u \in \mathcal{F}$.
- (5) For any $p, q \in X$,

$$\inf \left\{ \frac{\mathcal{E}(u)}{|u(p) - u(q)|^2} : u \in \mathcal{F}, u(p) \neq u(q) \right\} > 0.$$

- (6) If $u \in \mathcal{F}$ then $\bar{u} \in \mathcal{F}$ and $\mathcal{E}(\bar{u}) \leq \mathcal{E}(u)$ (the *Markov* property).

Remark 5.2.

- (1) The positive quantity on the left side of (5) will define $R(p, q)^{-1}$, where R is the resistance metric on X associated to the resistance form $(\mathcal{E}, \mathcal{F})$.
- (2) If X is finite then $(\mathcal{E}, \ell(X))$ is a resistance form iff $(\mathcal{E}, \ell(X))$ is a Dirichlet form as in Definition 3.2. In this case the expression in (5) is positive by the parenthetical remark following Definition 3.10.
- (3) If $\mathcal{S} = \{(V_m, H_m)\}$ is a compatible sequence then $(\mathcal{E}_{\mathcal{S}}, \mathcal{F}(\mathcal{S}))$ is a resistance form on V_* by Lemma 4.5 and Proposition 4.6.

$\square$

Definition 5.3 (Resistance metric). Suppose X is a set and $R : X \times X \rightarrow [0, \infty)$. Then R is a *resistance metric* if for any finite $V \subset X$ there is a Laplacian H_V on V such that $R|_{V \times V} = R_{H_V}$, where R_{H_V} is as in Definition 3.10.

Remark 5.4.

- (1) R is a metric on X since for any finite $V \subset X$, R_{H_V} is a metric on V by Theorem 3.13.
- (2) From Theorem 3.13:
 - (a) For any finite $V \subset X$, the Laplacian H_V is uniquely determined by R .
 - (b) If $V_1 \subset V_2 \subset X$ where both V_1 and V_2 are finite then $(V_1, H_{V_1}) \preceq (V_2, H_{V_2})$.
 - (c) If X is finite then R is the effective resistance corresponding to H_X .

- (3) If $\mathcal{S} = \{(V_m, H_m)\}$ is a compatible sequence then the resistance metric $R_{\mathcal{S}}$ on V_* is a resistance metric in the sense of the above Definition. (This follows by taking, for any finite $V \subset V_*$, a sufficiently large m such that $V \subset V_m$.) $\square$

Theorem 5.5. *Suppose $(\mathcal{E}, \mathcal{F})$ is a resistance form on a set X .*

Then for any finite $V \subset X$ and any $u \in \ell(V)$ there is an extension of u to $h_V(u) \in \mathcal{F}$ such that

$$(2) \quad \mathcal{E}(h_V(u)) = \min\{\mathcal{E}(v) : v \in \mathcal{F}, v|_V = u\}.$$

The map h_V is linear and $h_V(u)$ is the unique minimiser.

Moreover, $\mathcal{E}^V(u) := \mathcal{E}(h_V(u))$ defines a Dirichlet form $(\mathcal{E}^V, \ell(V))$ on $\ell(V)$ (as in Definition 3.2) and hence a resistance form (as noted in Remark 5.4), called the form induced from $\mathcal{E}$.

If $V_1 \subset V_2 \subset X$ where both V_1 and V_2 are finite then the form induced on V_1 from $\mathcal{E}^{V_2}$ is the same as the form induced from $\mathcal{E}$.

Define

$$(3) \quad R_{p,q}^{-1} := \min\{\mathcal{E}(v) : v \in \mathcal{F}, v(p) = 1, v(q) = 0\}.$$

Then $R_{p,q}$ defines a resistance metric on X . It agrees with the effective resistance given by Definition 3.10 on any finite $V \subset X$ with $p, q \in V$.

Proof. Suppose $V \subset X$ is finite and $u \in \ell(V)$. Suppose $h_V(u), v \in \mathcal{F}$ agree with u on V , and let $v = h_V(u) + \tilde{v}$, in which case $\tilde{v} \in \mathcal{F}$ and $\tilde{v} = 0$ on V . Then

$$(4) \quad \mathcal{E}(v) = \mathcal{E}(h_V(u)) + 2\mathcal{E}(h_V(u), \tilde{v}) + \mathcal{E}(\tilde{v}),$$

Thus in order to obtain (2) we need to define $h_V(u)$ agreeing with u on V and so that $\mathcal{E}(h_V(u), \tilde{v}) = 0$ for all $\tilde{v} \in \mathcal{F}$ with $\tilde{v} = 0$ on V .

For this, first fix $p \in V$ and consider $\mathcal{X}_p \in \ell(V)$, where $\mathcal{X}_p$ is the characteristic function of the singleton $\{p\}$. Let

$$\mathcal{F}^p = \{v \in \mathcal{F} \mid v = 0 \text{ on } V \setminus \{p\}\}.$$

Then $(\mathcal{F}^p, \mathcal{E})$ is non trivial by Definition 5.1(4) and is a Hilbert space by Definition 5.1(3) and (5). (Use (5) to show that $\mathcal{E}(v_n - v) \rightarrow 0$ and $v_n(q) = 0$ implies $v(q) = 0$.)

Moreover, $p \mapsto v(p)$ is a linear map on $\mathcal{F}^p$ and is bounded in the $\mathcal{E}$ norm by Definition 5.1(5) after fixing some $q \in V \setminus \{p\}$. Hence there exists $g_p \in \mathcal{F}^p$ such that

$$\mathcal{E}(g_p, v) = v(p) \quad \forall v \in \mathcal{F}^p.$$

Define the function $h_V(u) \in \mathcal{F}$ by

$$h_V(u) = \sum_{p \in V} u(p) \frac{g_p}{g_p(p)}.$$

Then $h_V(u) = u$ on V . Moreover, if $\tilde{v} \in \mathcal{F}$ and $\tilde{v} = 0$ on V , then

$$\mathcal{E}(h_V(u), \tilde{v}) = \sum_{p \in V} \frac{u(p)}{g_p(p)} \mathcal{E}(g_p, \tilde{v}) = \sum_{p \in V} \frac{u(p)}{g_p(p)} \tilde{v}(p) = 0.$$

Using the remark following (4) it follows that if $v = u$ on V then $\mathcal{E}(h_V(u)) \leq \mathcal{E}(v)$. Moreover, by Definition 5.1(3) equality holds by iff $\tilde{v}$ is constant, i.e. $\tilde{v} = 0$ and so $h_V(u) = v$. Hence $h_V(u)$ is the unique energy minimising extension of u .

The form $\mathcal{E}^V$ is clearly symmetric, bilinear and positive definite. Next note that $h_V(u)$ is constant iff u is constant, one direction being trivial and the other using the fact that the constant extension of a constant u minimises energy (being zero) and so is the unique extension $h_V(u)$. Hence the first condition in Definition 3.2 is satisfied.

Next note

$$\mathcal{E}(h_V(\bar{u})) \leq \mathcal{E}(\overline{h_V(u)}) \leq \mathcal{E}(h_V(u)).$$

The first inequality is since $u = h_V(u)$ on V implies $\bar{u} = \overline{h_V(u)}$ on V , and by then applying the energy minimisation property of h_V . The second inequality is by the Markov property of $\mathcal{E}$.

It follows $\mathcal{E}^V(\bar{u}) \leq \mathcal{E}^V(u)$. This is the Markov property for $\mathcal{E}^V$.

Hence $\mathcal{E}^V$ is a Dirichlet form on V as in Definition 3.2.

Let $\mathcal{E}^{V_1 \rightarrow V_2}$ be the form induced on V_1 from $\mathcal{E}^{V_2}$. Then for $u \in \ell(V_1)$,

$$\begin{aligned} \mathcal{E}^{V_1 \rightarrow V_2}(u) &= \min \{ \mathcal{E}^{V_2}(v) \mid v \in \ell(V_2), v|_{V_1} = u \} \\ &= \min \left\{ \min \{ \mathcal{E}^X(v') \mid v' \in \mathcal{F}, v'|_{V_2} = v \} \mid v \in \ell(V_2), v|_{V_1} = u \right\} \\ &= \min \{ \mathcal{E}^X(v') \mid v' \in \mathcal{F}, v'|_{V_1} = u \} \\ &= \mathcal{E}^{V_1}(u). \end{aligned}$$

The third equality is trivial, we are just splitting up the class of extensions into a union of disjoint subclasses. Hence $\mathcal{E}^{V_1 \rightarrow V_2} = \mathcal{E}^{V_1}$.

Finally, if $p, q \in V \subset X$ for any finite V , then from what we have just shown with $V_1 = \{p, q\}$ and $V_2 = V$, it follows that $R_{p,q}$ is the effective resistance between p and q w.r.t. the Dirichlet form $\mathcal{E}^V$ induced from $\mathcal{E}$. If H_V is the Laplacian on V corresponding to $\mathcal{E}^V$ then $R_{p,q} = R_{H_V}(p, q)$ from Definition 3.10. It follows $R_{p,q}$ defines a resistance metric on X . $\square$

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